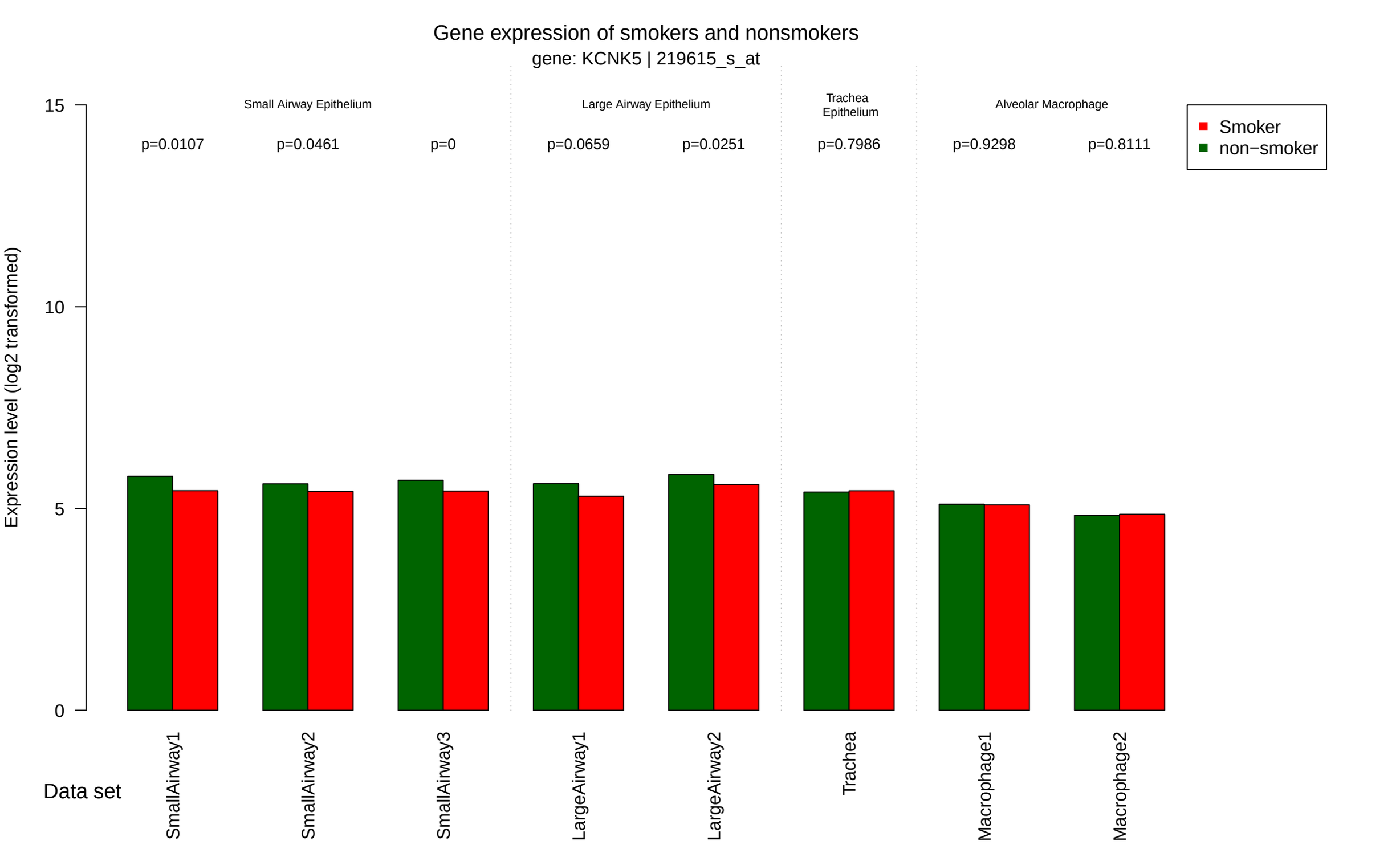
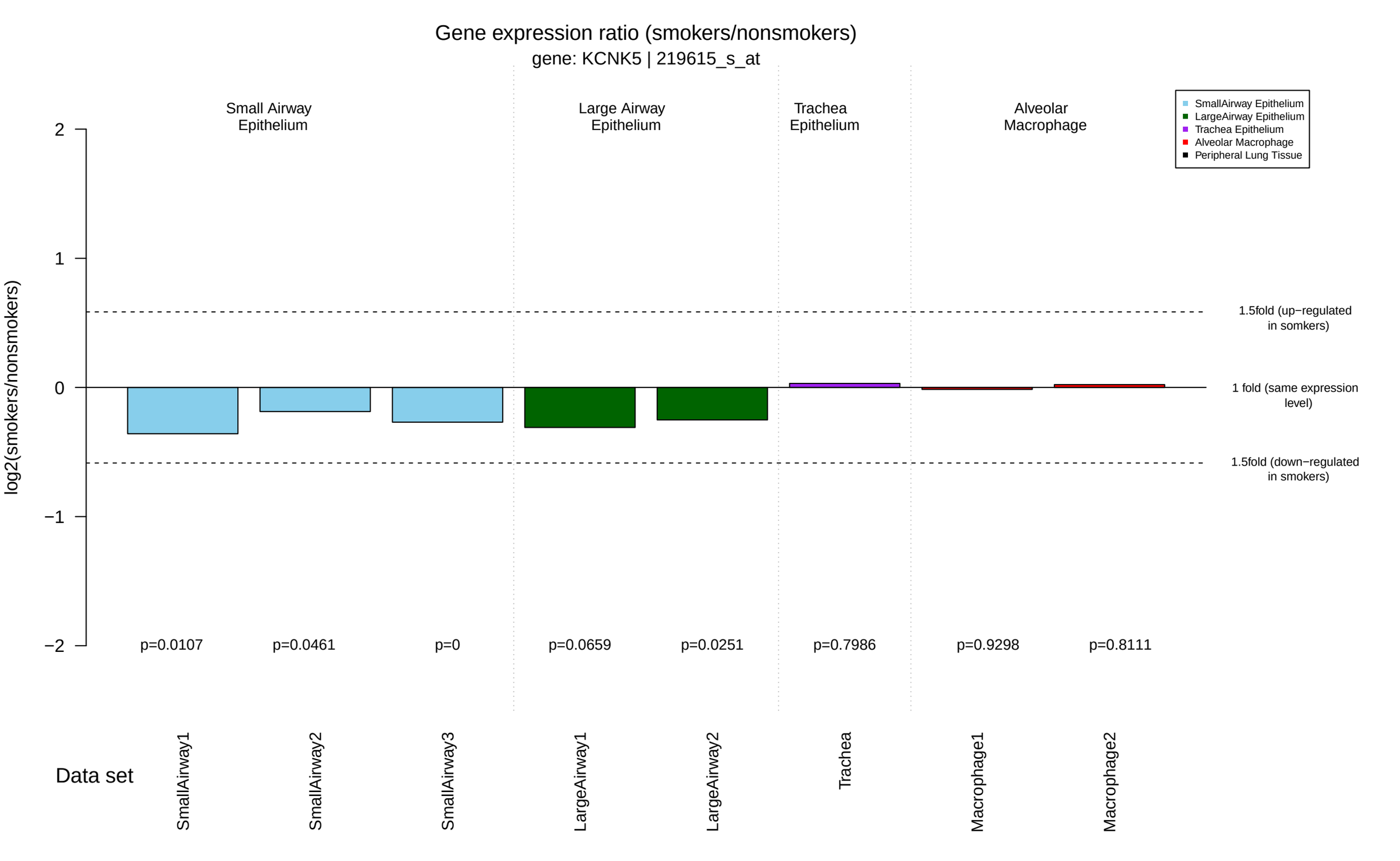
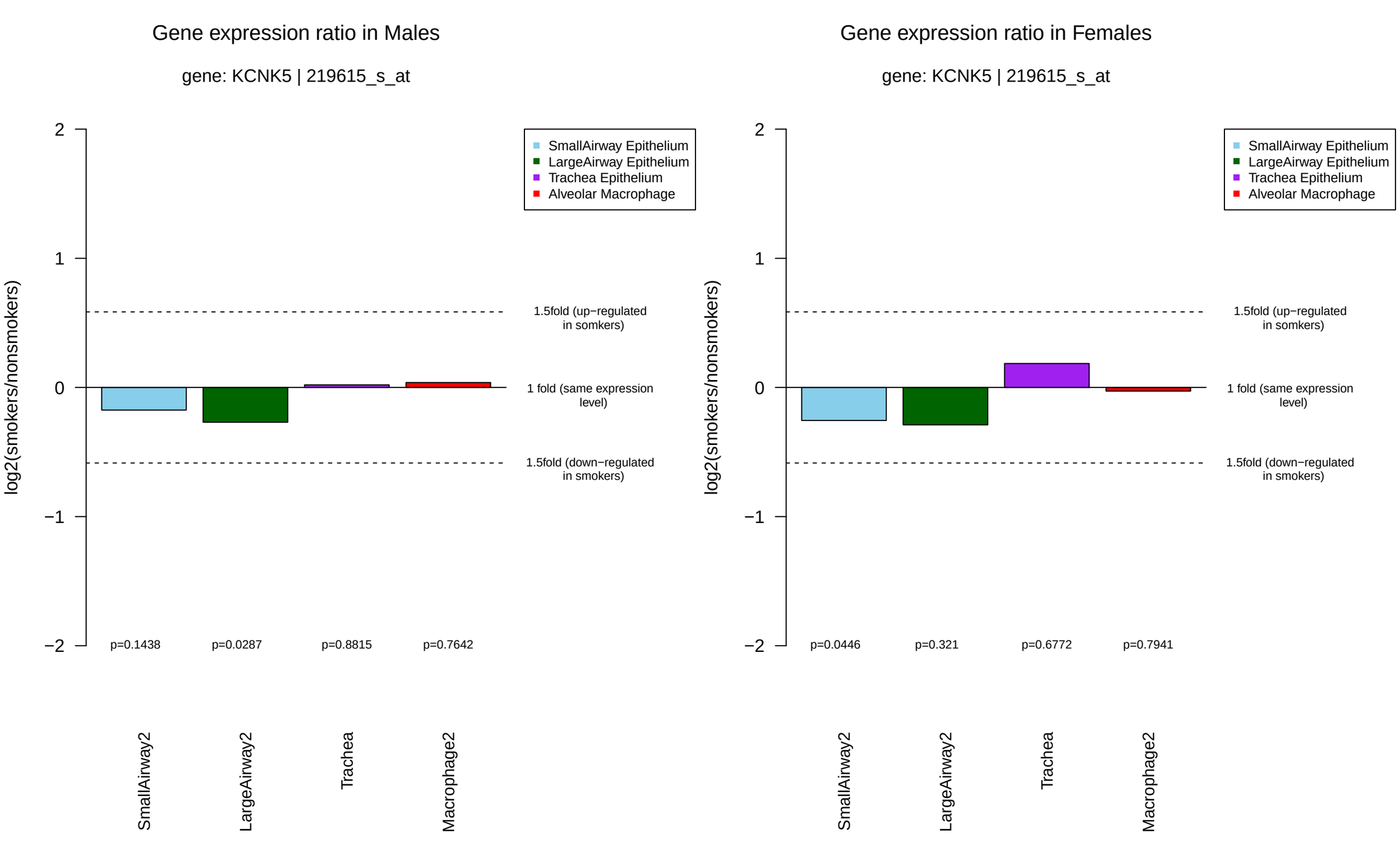
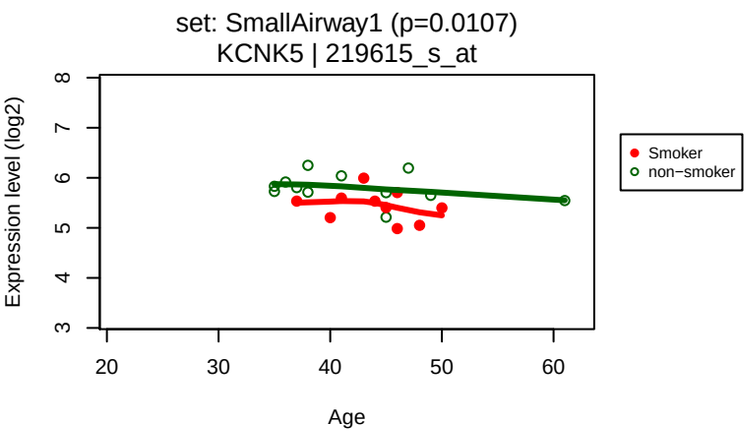
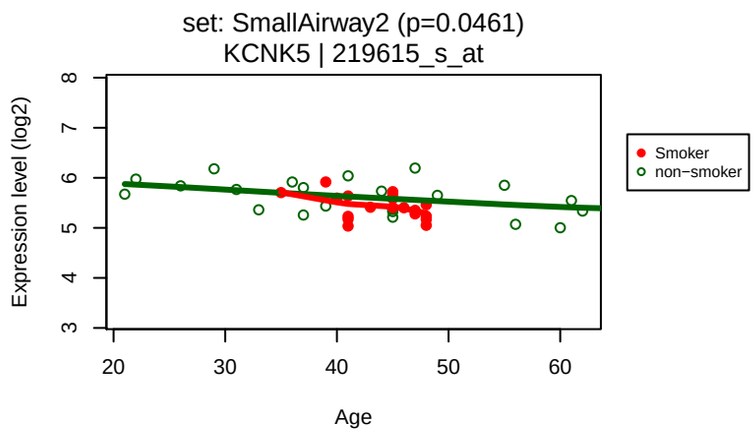
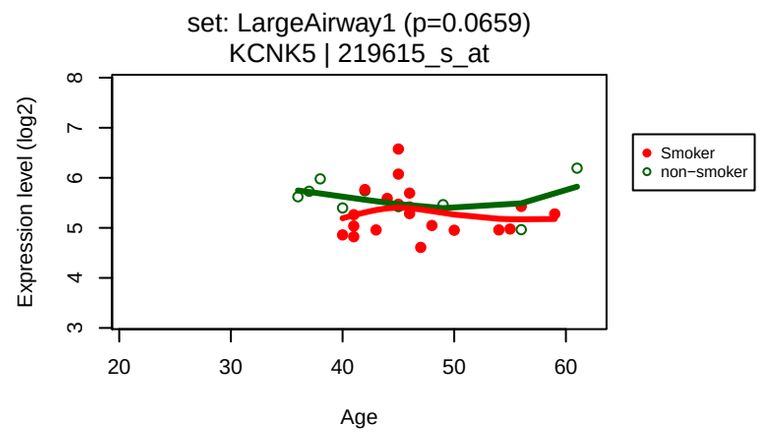

Gene expression of smokers and nonsmokers
gene: KCNK5 | 219615_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0107
p=0.0461
p=0
p=0.0659
p=0.0251
p=0.7986
p=0.9298
p=0.8111
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

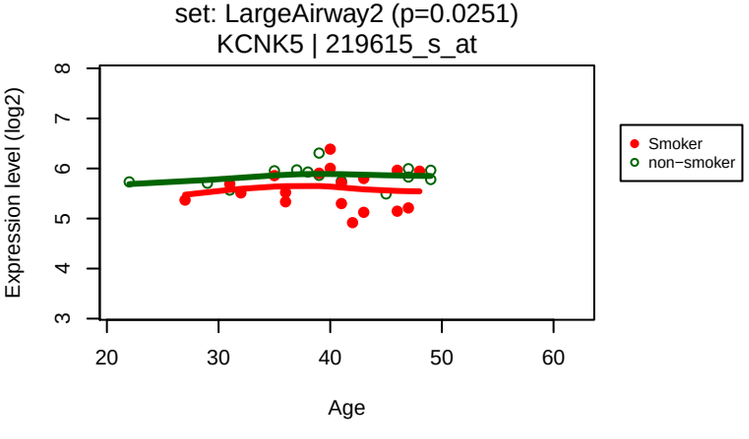

Gene expression ratio (smokers/nonsmokers)
gene: KCNK5 | 219615_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0107
p=0.0461
p=0
p=0.0659
p=0.0251
p=0.7986
p=0.9298
p=0.8111
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

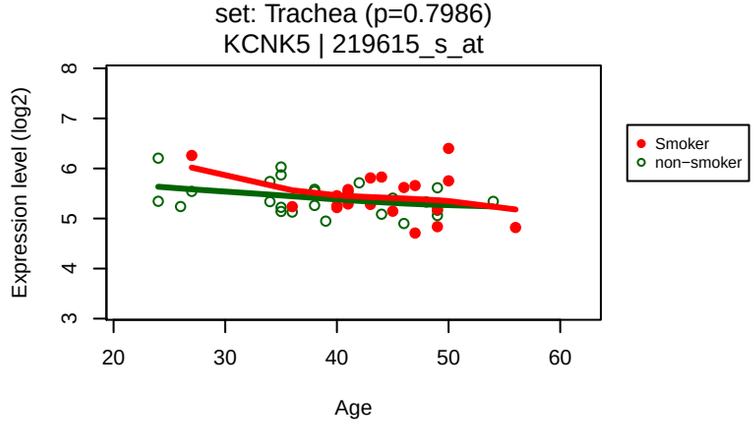

Gene expression ratio in Males
gene: KCNK5 | 219615_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1438
p=0.0287
p=0.8815
p=0.7642
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KCNK5 | 219615_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0446
p=0.321
p=0.6772
p=0.7941
SmallAirway2
LargeAirway2
Trachea
Macrophage2

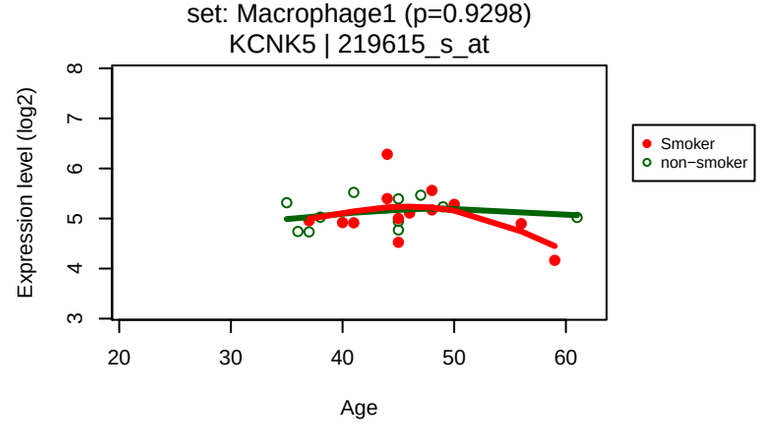

set: SmallAirway1 (p=0.0107)
KCNK5 | 219615_s_at
Expression level (log2)
Age
Smoker
non−smoker

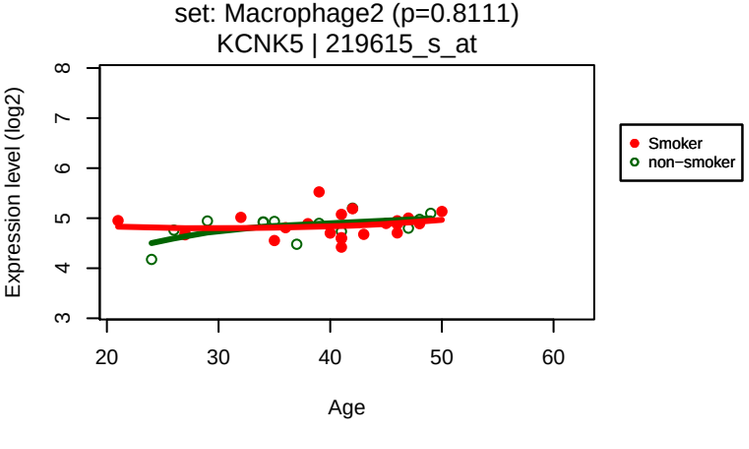

set: SmallAirway2 (p=0.0461)
KCNK5 | 219615_s_at
Expression level (log2)
Age
Smoker
non−smoker

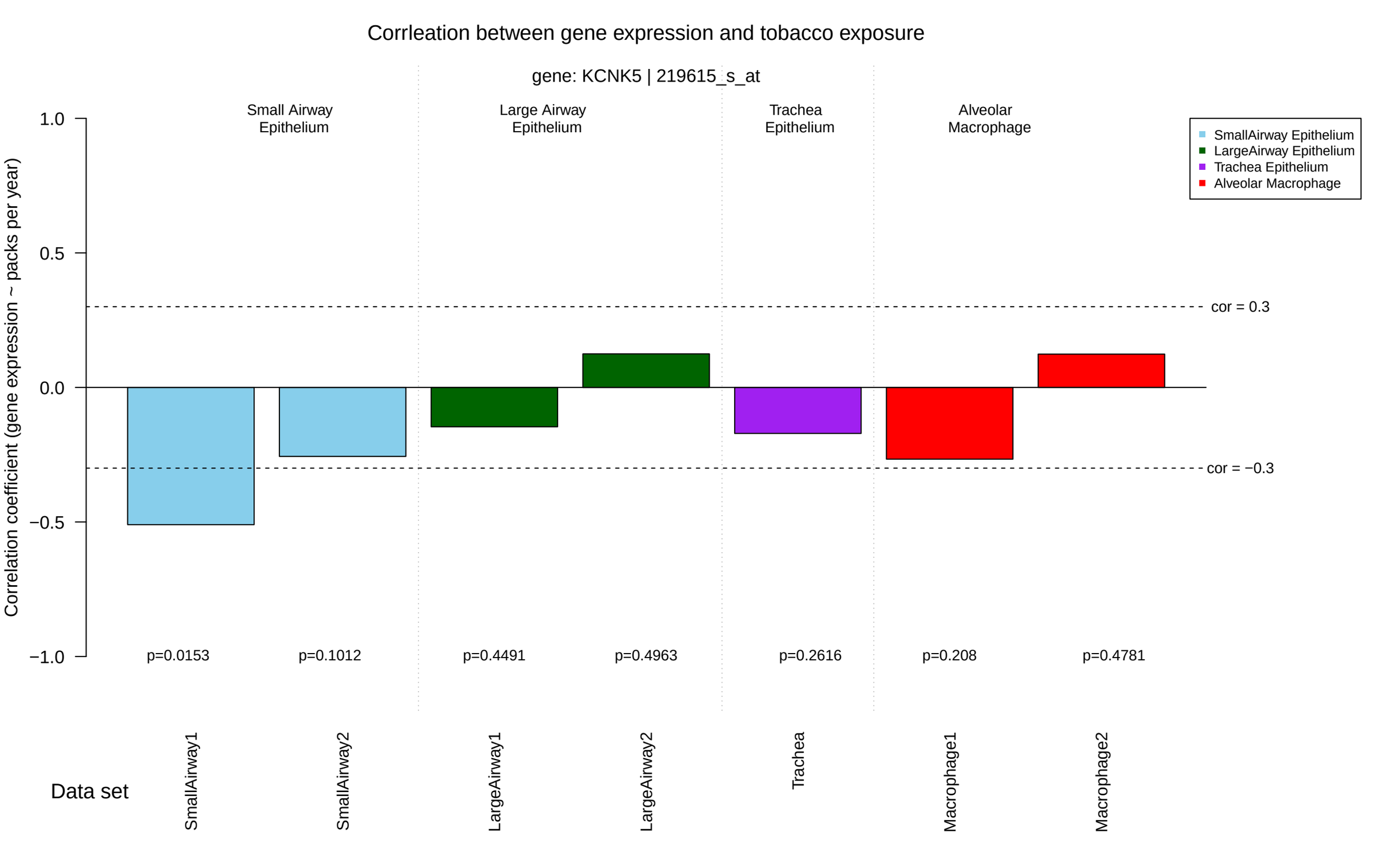

set: LargeAirway1 (p=0.0659)
KCNK5 | 219615_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0251)
KCNK5 | 219615_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7986)
KCNK5 | 219615_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9298)
KCNK5 | 219615_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8111)
KCNK5 | 219615_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KCNK5 | 219615_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0153
p=0.1012
p=0.4491
p=0.4963
p=0.2616
p=0.208
p=0.4781
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2